

Element H:

H1. Addressing Requirements

In order to gather data on our design requirements, we asked some questions:

- Does the desk provide adequate room for Ms. A.V 's work documents and sentimental items?
- Does the desk have enough space for Ms. A.V 's monitor, computer and outlet?
- Is the desk aesthetically pleasing to the stakeholder?
- Can we afford the materials / are the materials easily accessible to us?

We plan to test how well our desk functions by evaluating how well the desk serves Ms. A.V 's work needs, how aesthetically pleasing it is, how durable and functional it is, and overall make sure that the prototype has structural integrity and aesthetic in order for it to be effective for MS.A.V in all aspects when it comes to productivity and ergonomics.

Expected Results (s):

H2. Conduct of Tests

- **Stability Test:** With a stability test we can make sure what our model needs in order to be stable. When we test our model we can see potential areas where our desk might break apart. With this test we can reinforce areas and take notes on where the most problems may occur when we begin to build our actual design.
 - Setup: Taking the prototype model of the desk and placing weight on the desk to see how well the legs withstand the weight
 - Expected Range: One or two issues with the legs of the desk, particularly where they connect to the top, but should be able to withstand the weight
 - Minimum Goal: Able to withstand the weight but might be shaky
 - Goal: The prototype is able to hold the weight.
- **Aesthetics Test:** With this test we can begin a survey for the design of the desk. We can ask Caylee's entire family as well as our mentors to rate the looks of the desk. We'll ask for detailed feedback and thoughts so that we are able to possibly implement them into our final project.
 - Setup: Mentors and Stakeholders will be given a questionnaire asking for their opinions on the desk
 - Expected Range: Few concerns and questions about the different aesthetics of the desk such as flower petals

- Minimum Goal:
- Goal: No more than 5 concerns about the design.
- **Durability Test:** During this test, we will expose the prototype to the normal events that will occur when it's being used, such as having items placed on top of the desk and storing items within the desk.
 - Setup: Will pla
 - Expected Range: Should be able to do all the functions of a normal desk
 - Minimum Result:
 - Goal: No damage after 15 tests

H3. Quantity of Tests Planned

We will do a sufficient amount to cover all the important needs, requirements, and criteria for the prototype. These tests include:

- Test how aesthetic and appealing to the eye the desk is.
- Test its structural integrity to see how durable it is to withstand unexpected events when it is exposed to it.
- Test its stability and make sure that parts that could be known as fragile won't break and withstand any events or weight that it will face when being used.
- Overall test its functionality and effectiveness for our stakeholder Ms.A.V .

Each test area will be done at least 10 trials to make sure the results are accurate and not random. We want to make sure that we have a wide range of data to make sure that we can analyze all aspects and flaws of the desk as much as we can to ensure that we can make changes and improvements for our final prototype and maximize its functionality and effectiveness. We will make sure we don't just do the same type of tests. We will try to have different conditions, situations, or circumstances in each test so that our prototype can have exposure to many different types of unexpected situations that could happen in life. Furthermore, this will allow us to get a wide range of data and overall analyze better our prototype so that we can use the knowledge to create a final prototype where its effectiveness and functionality is maximized to its best version.

H4.Logic of the plan

Our test is lociagal because we will test certain topics that will benefit our design. We will test eac need/requirement carefully and consistently do it so we can get accurate results which will help us make improvements in the future for our final prototype.

- **Stability:** We will test stability by using multiple different weights/items to check how much our design can hold.
- **Aesthetic:** We can test the aesthetic part of our design by sharing a survey with our stakeholders to get their opinion of the design.
- **Durability:** We can test that the desk is very functional and durable by testing to see how much weight it can hold, if it damages easily if water gets on it, if we shake the desk, hit it with some force, or throw it with some force. Furthermore, tests like these will allow us to analyze how well our desk can withstand circumstances in certain situations.

The tests are designed to keep everything controlled, and we will follow the right methods to get accurate results. We will make sure that we compare the results of each test for the desk before and after the improvements that were made. As well, each test will have at least 10 trials to make sure that we get a wide range of data and make sure that the results are accurate and reliable that we can use to infer and make improvements on our prototype.

H5. Use of field experts

We have asked three experts who would be knowledgeable about how to build a desk to get reasonable information that would benefit our building process.

- **Mr. R.N (Engineering Teacher/Founder of Construction Club):** He gave ideas on how to design the resin of our build so that it would be aesthetic and some other ideas on what we could change on the structural integrity of the desk itself in order to ensure that it's both durable and stable withstand any situations when the desk is in use by Ms.A.V .
- **Mr. T.B(Mechanical Engineer/Woodworker):** He gave advice on how to build our design, especially focused on the structural integrity of the desk and pull-out drawers. We are also about materials we can use that are effective, but also affordable. Finally, he gave us ideas on the tests we can do to test the needs and make sure the prototype meets the requirements.
- **Mr. L.P (Engineering Teacher)** – He gave us advice and feedback over the scaled version of our desk prototype and gave us advice on what we areas we can improve on when it comes to building our final prototype and how we can build our final prototype such as giving us advice of new ideas for improvements, materials, etc..

All our field experts looked at both our prototype and testing methods for it where they gave us good ideas on how to test and gave suggestions on how to improve it. They gave us valuable feedback where for example we got a suggestion on how we could keep it stable under an uneven surface by adding adjustable caps underneath.

In conclusion, All these tests will help with our final prototype/ design for our desk. It makes sure that Mrs. Vasques has all the needs and wants with our final design so that she can have a comfortable desk to work on. By focusing on the important design needs, doing thorough tests, and listening to experts, we are confident that our solution will help Ms.A.V with the lack of productivity, workspace, and ergonomics. Furthermore, we are sure our prototype of our desk will be effective and functional for Ms.A.V and overall satisfy her and improve her ergonomics.